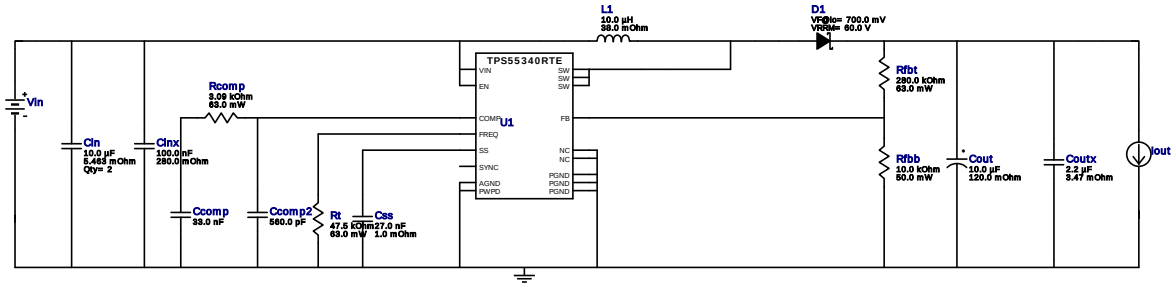
WEBENCH ® Design Report

VinMin = 11.0V
VinMax = 12.0V
Vout = 36.0V
Iout = 0.7A

Device = TPS55340RTER
Topology = Boost
Created = 2026-07-11 00:09:28.624
BOM Cost = $2.54
BOM Count = 15
Total Pd = 1.23W

Design : 28 TPS55340RTER
TPS55340RTER 10V-12V to 32.00V @ 0.7A



1. For frequency > 1.2 MHz, the user is required to make sure load current is always greater than "Iout Min" value shown in the Op Vals.

Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Ccomp	TDK	CGA4J2C0G1H333J125AA Series= C0G/NP0	Cap= 33.0 nF VDC= 50.0 V IRMS= 0.0 A	1	$0.08	0805 7 mm²
Ccomp2	MuRata	GRM1555C1H561JA01J Series= C0G/NP0	Cap= 560.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Cin	Taiyo Yuden	MSAST21GBB5106MTNA01 Series= X5R	Cap= 10.0 uF ESR= 5.463 mOhm VDC= 25.0 V IRMS= 2.36768 A	2	$0.04	0805 7 mm²
Cinx	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Cout	Panasonic	EEHZA1H100R Series= ZA	Cap= 10.0 uF ESR= 120.0 mOhm VDC= 50.0 V IRMS= 750.0 mA	1	$0.46	SM_RADIAL_5MM 58 mm²
Coutx	TDK	C2012X5R1H225K125AB Series= X5R	Cap= 2.2 uF ESR= 3.47 mOhm VDC= 50.0 V IRMS= 0.0 A	1	$0.13	0805 7 mm²
Css	MuRata	GRM155R71A273KA01D Series= X7R	Cap= 27.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
D1	Diodes Inc.	B560C-13-F	VF@Io= 700.0 mV VRRM= 60.0 V	1	$0.19	SMC 83 mm²
L1	NIC Components	NPI31W100MTRF	L= 10.0 µH 38.0 mOhm	1	$0.36	IND_NPI31W 172 mm²
Rcomp	Vishay-Dale	CRCW04023K09FKED Series= CRCW..e3	Res= 3.09 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²

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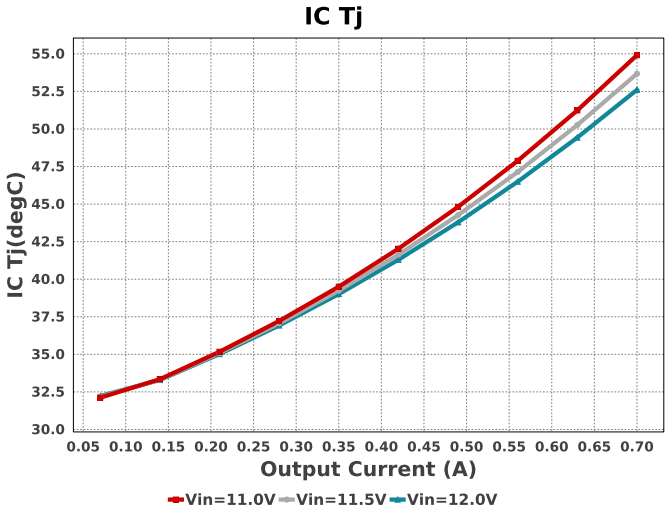
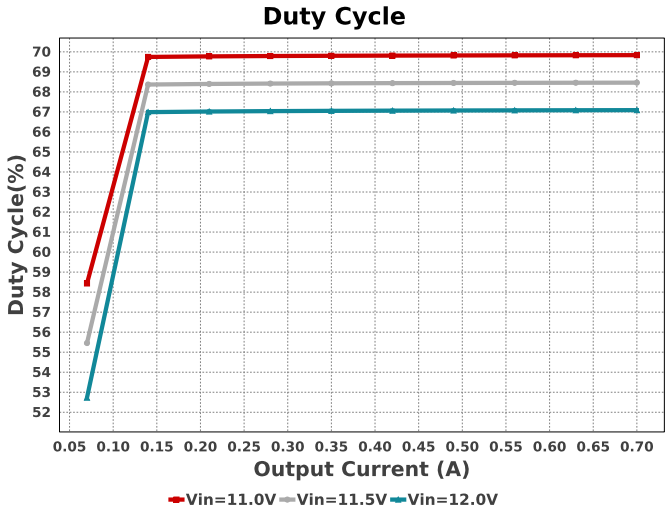
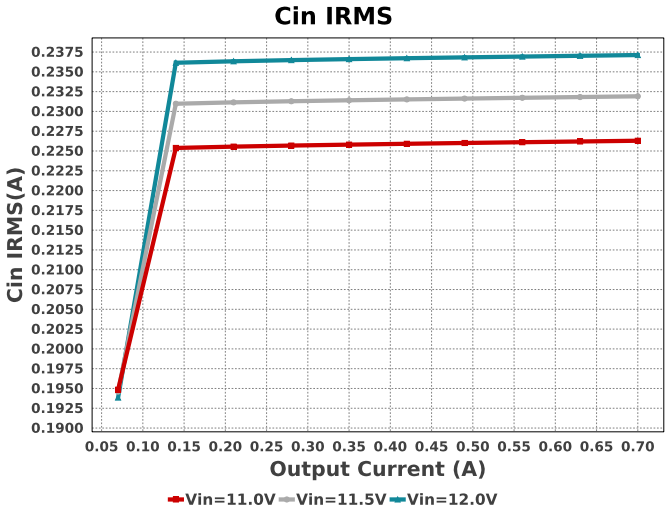
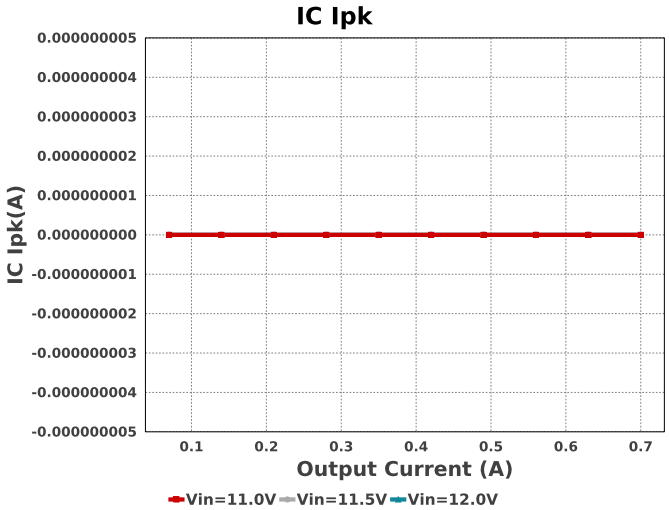
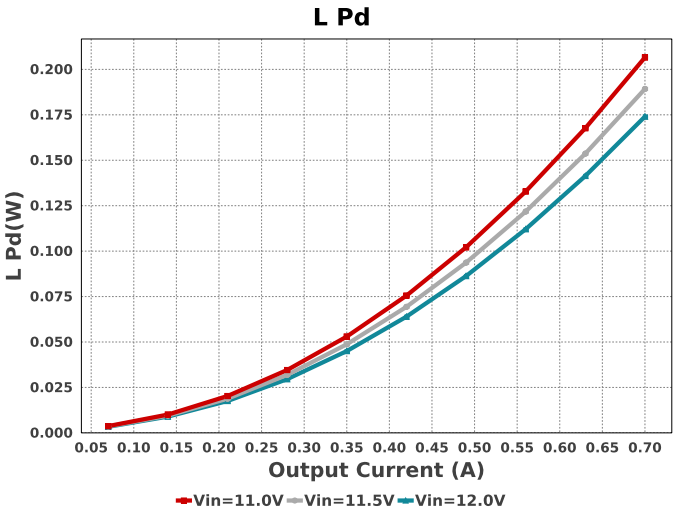
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rfbb	Yageo	RC0201FR-0710KL Series= ?	Res= 10.0 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	0201 2 mm²
Rfbt	Vishay-Dale	CRCW0402280KFKED Series= CRCW..e3	Res= 280.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rt	Vishay-Dale	CRCW040247K5FKED Series= CRCW..e3	Res= 47.5 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
U1	Texas Instruments	TPS55340RTER	Switcher	1	$1.17	RTE0016C 16 mm²

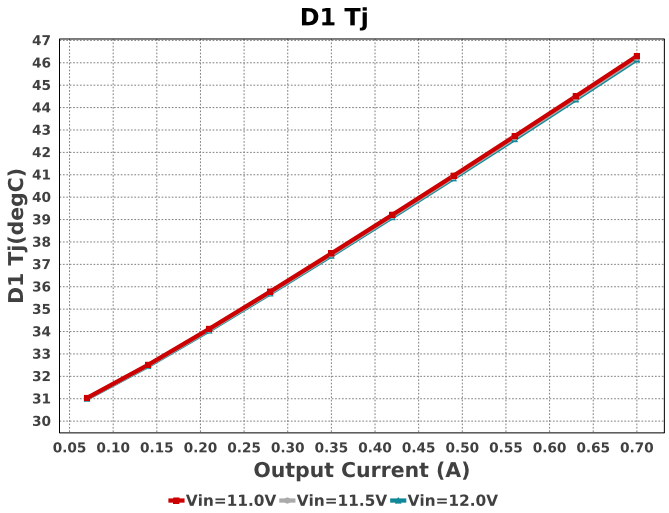


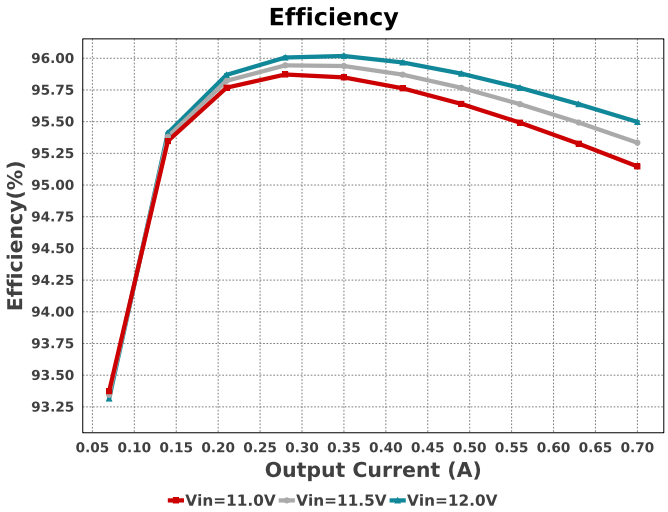


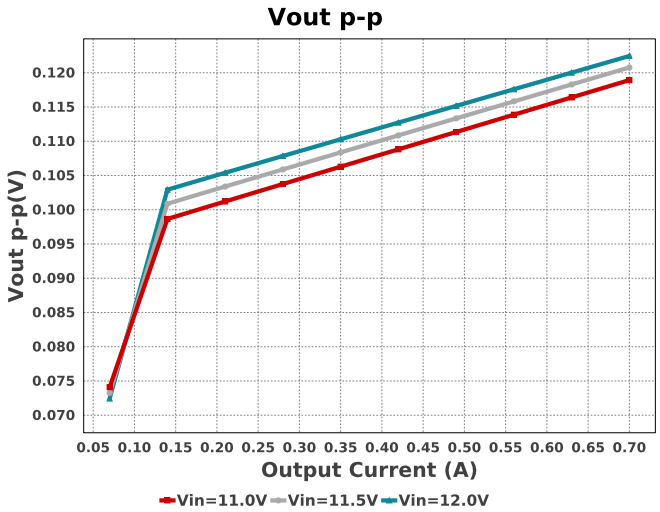


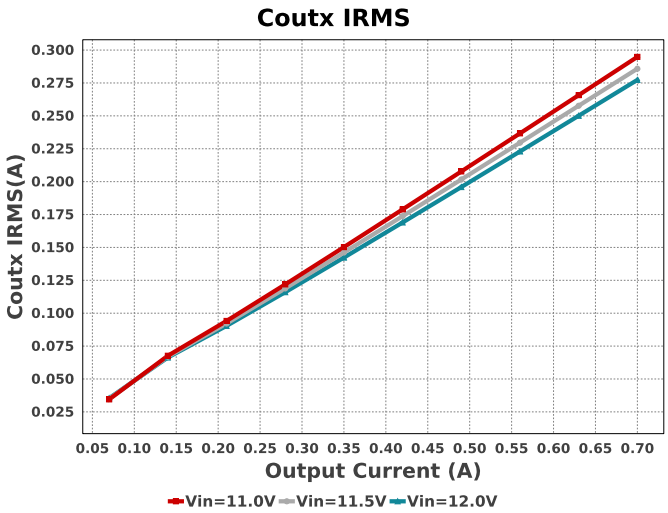


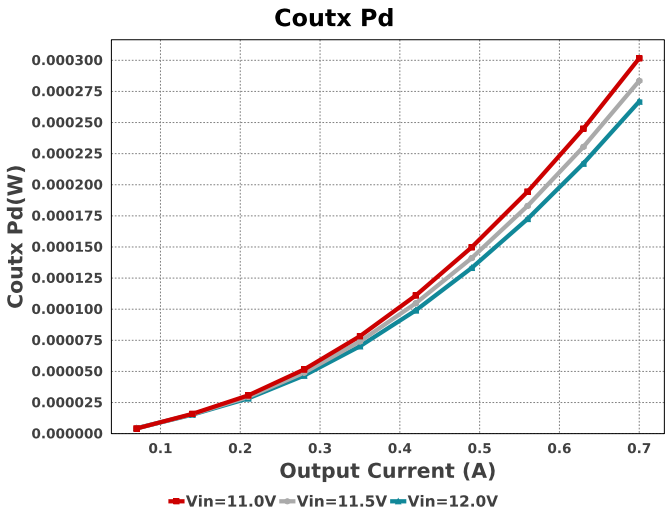












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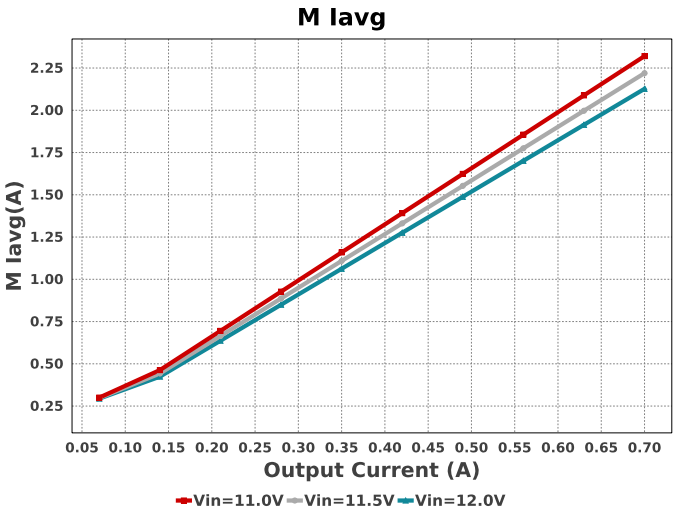

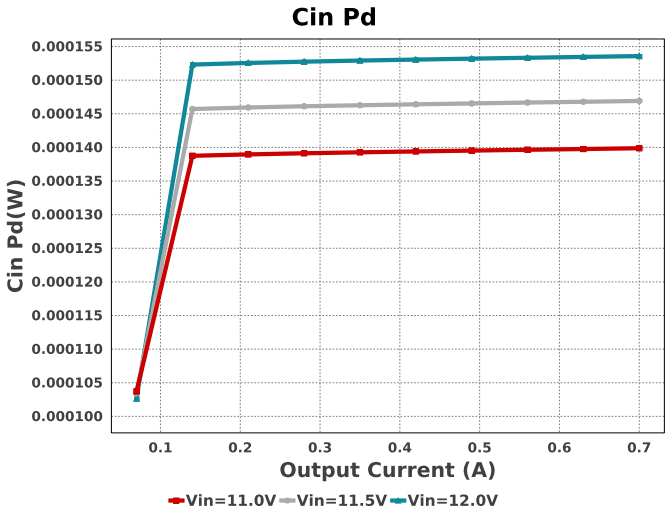

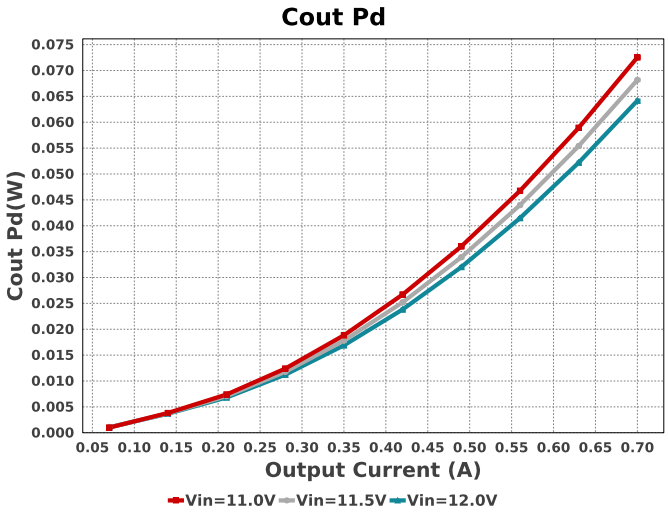

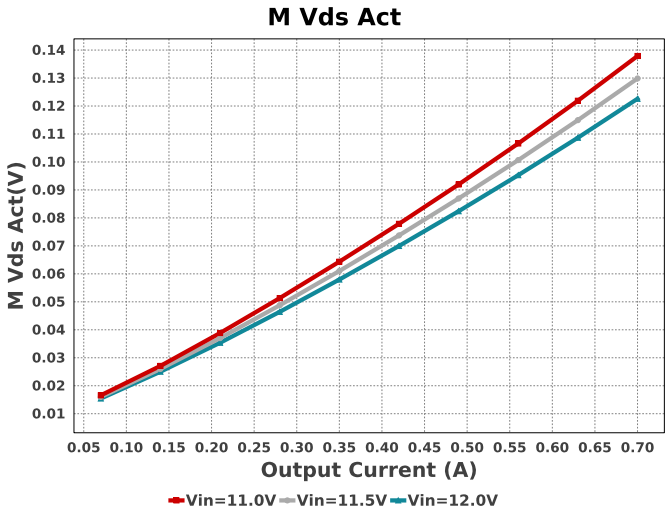

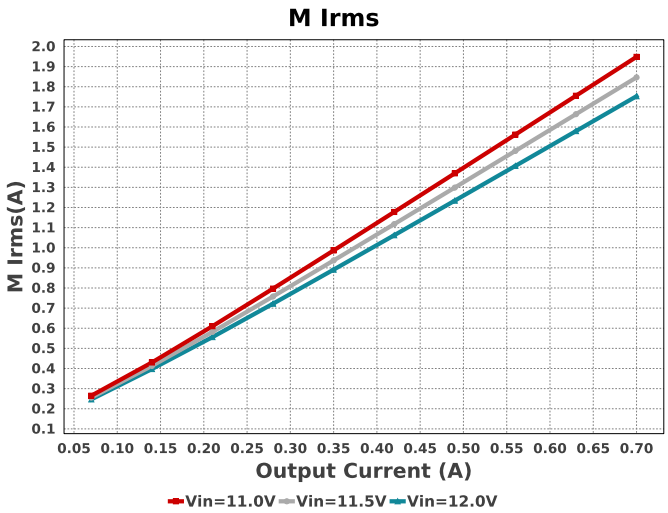

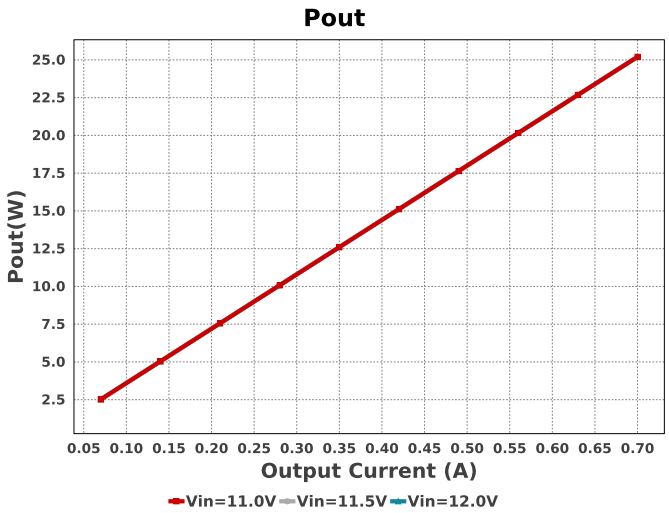

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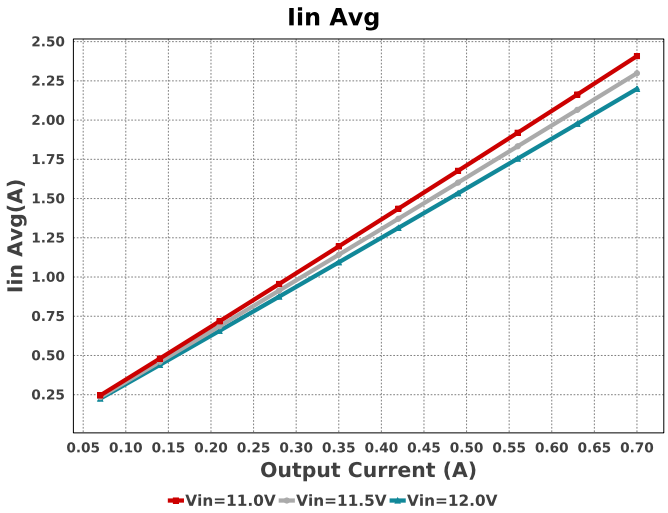

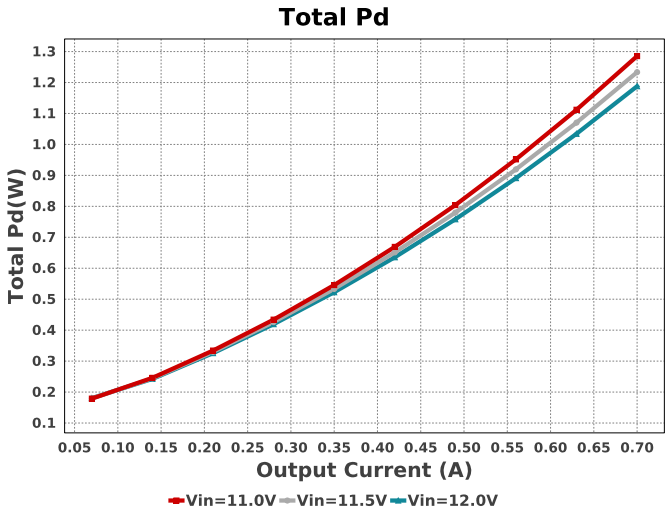

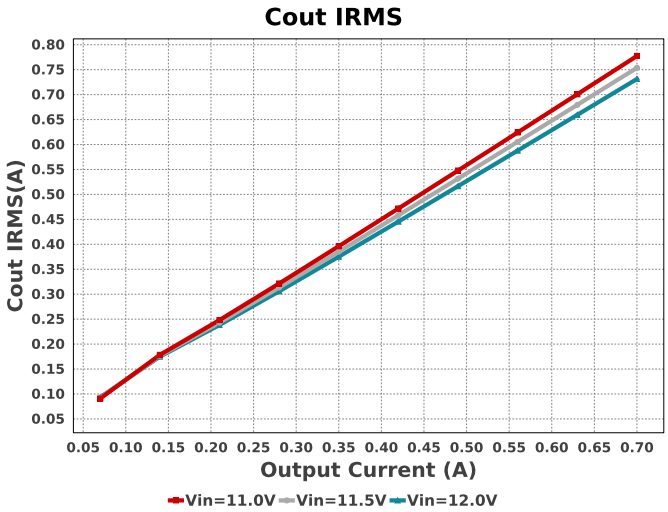

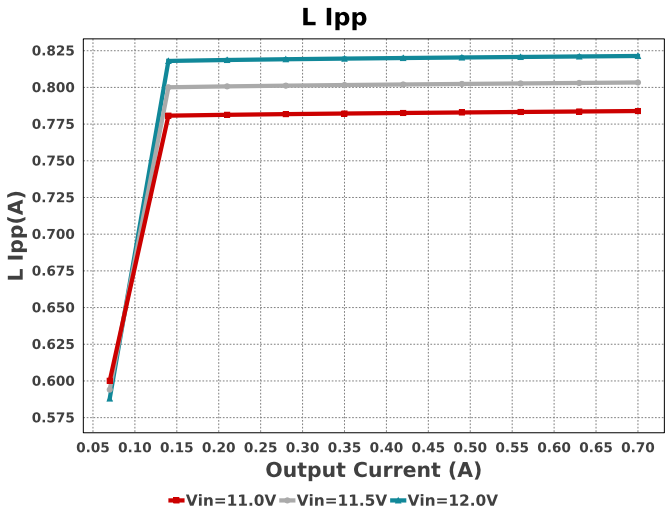

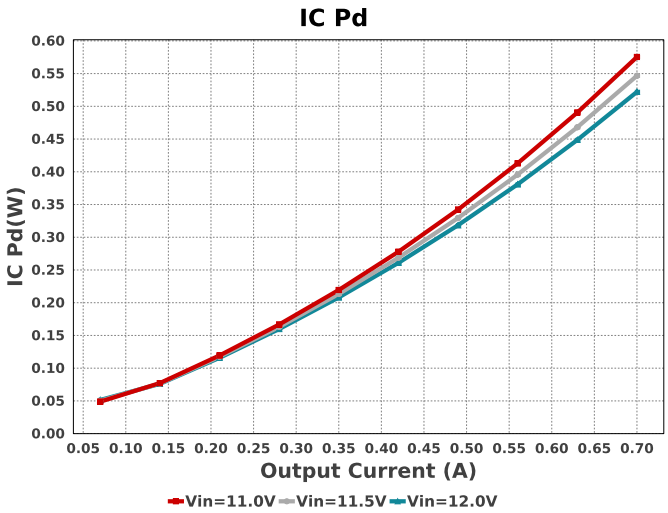

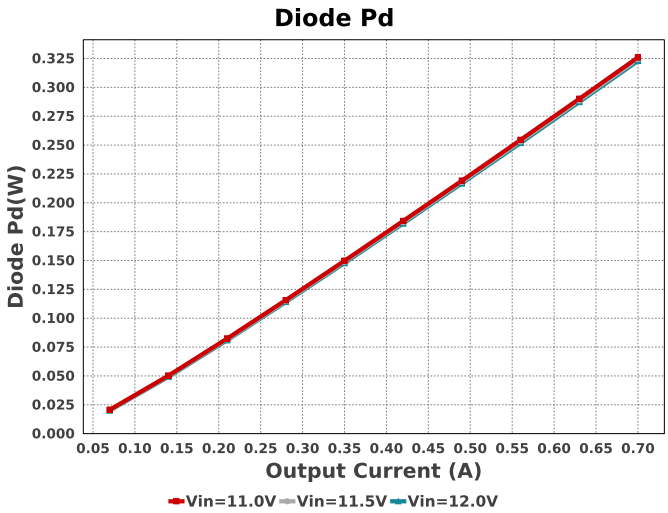

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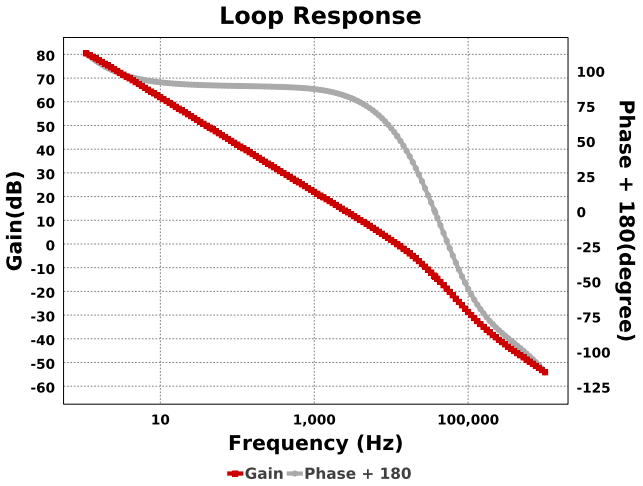

Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	226.08 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	139.61 µW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	776.253 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	72.308 mW	Capacitor	Output capacitor power dissipation
5.	Coutx IRMS	294.381 mA	Capacitor	Output capacitor_x RMS ripple current
6.	Coutx Pd	300.71 µW	Capacitor	Output capacitor_x power loss
7.	D1 Tj	43.49 degC	Diode	D1 junction temperature
8.	Diode Pd	269.8 mW	Diode	Diode power dissipation
9.	IC Ipk	0.0 A	IC	Peak switch current in IC
10.	IC Pd	573.14 mW	IC	IC power dissipation
11.	IC Tj	54.817 degC	IC	IC junction temperature
12.	IC Tolerance	9.0 mV	IC	IC Feedback Tolerance
13.	ICThetaJA	43.3 degC/W	IC	IC junction-to-ambient thermal resistance
14.	Iin Avg	2.402 A	IC	Average input current
15.	L Ipp	783.16 mA	Inductor	Peak-to-peak inductor ripple current
16.	L Pd	205.67 mW	Inductor	Inductor power dissipation
17.	M Iavg	2.315 A	Mosfet	MOSFET Average current
18.	M Irms	1.943 A	Mosfet	MOSFET RMS ripple current
19.	M Vds Act	137.452 mV	Mosfet	Voltage drop across the MosFET
20.	Cin Pd	139.61 µW	Power	Input capacitor power dissipation
21.	Cout Pd	72.308 mW	Power	Output capacitor power dissipation
22.	Coutx Pd	300.71 µW	Power	Output capacitor_x power loss
23.	Diode Pd	269.8 mW	Power	Diode power dissipation
24.	IC Pd	573.14 mW	Power	IC power dissipation
25.	L Pd	205.67 mW	Power	Inductor power dissipation
26.	Total Pd	1.226 W	Power	Total Power Dissipation
27.	BOM Count	15	System Information	Total Design BOM count
28.	Cross Freq	11.011 kHz	System Information	Bode plot crossover frequency
29.	Duty Cycle	69.768 %	System Information	Duty cycle
30.	Efficiency	95.361 %	System Information	Steady state efficiency
31.	FootPrint	381.0 mm^2	System Information	Total Foot Print Area of BOM components
32.	Frequency	983.333 kHz	System Information	Switching frequency
33.	Gain Marg	-13.276 dB	System Information	Bode Plot Gain Margin
34.	Iout	700.0 mA	System Information	Iout operating point
35.	Low Freq Gain	80.25 dB	System Information	Gain at 1Hz
36.	Mode	CCM	System Information	Conduction Mode
37.	Phase Marg	53.991 deg	System Information	Bode Plot Phase Margin
38.	Pout	25.2 W	System Information	Total output power
39.	Total BOM	$2.54	System Information	Total BOM Cost

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#	Name	Value	Category	Description
40.	Vin	11.0 V	System Information	Vin operating point
41.	Vout	36.0 V	System Information	Operational Output Voltage
42.	Vout Actual	35.641 V	System Information	Vout Actual calculated based on selected voltage divider resistors
43.	Vout Tolerance	2.697 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
44.	Vout p-p	118.812 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	700.0 m	Maximum Output Current
VinMax	12.0	Maximum input voltage
VinMin	11.0	Minimum input voltage
Vout	36.0	Output Voltage
base_pn	TPS55340	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

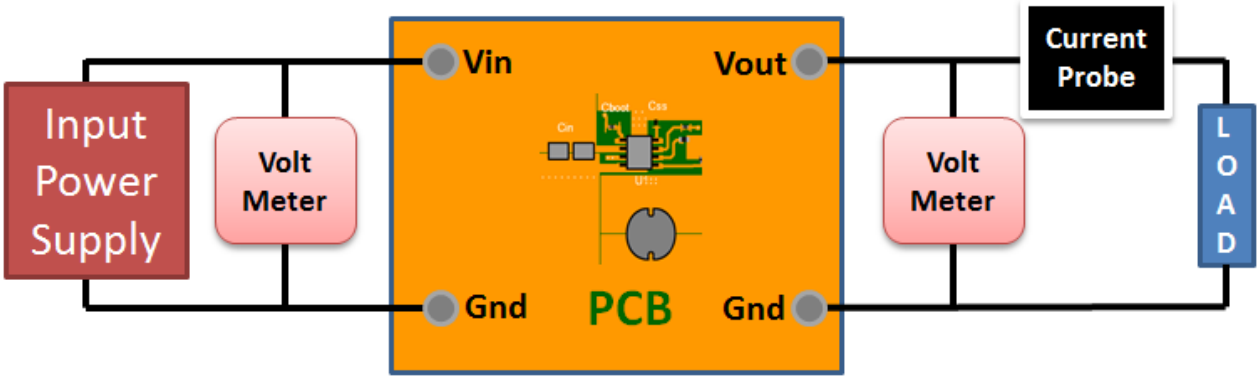
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 11.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



WEBENCH® Electrical Simulation Report

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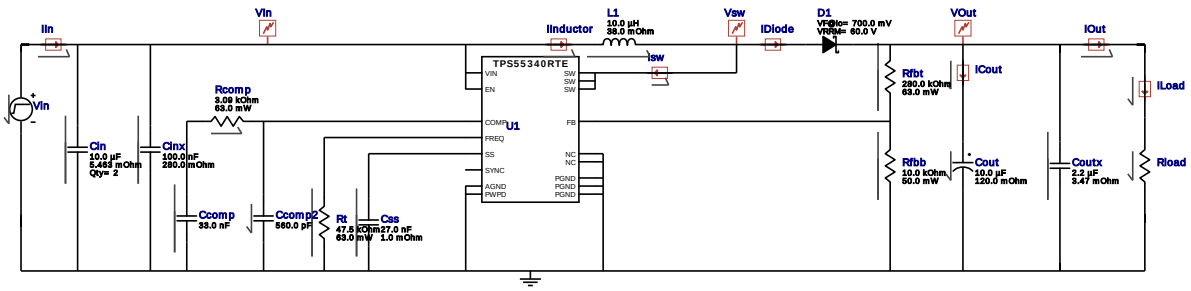
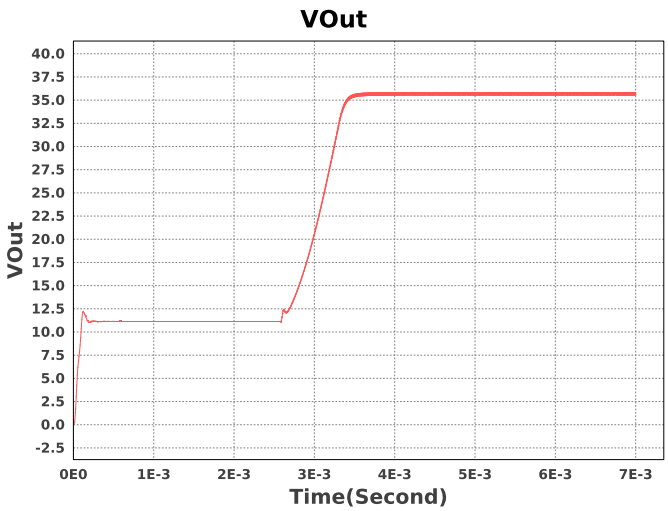
sim_id = 1

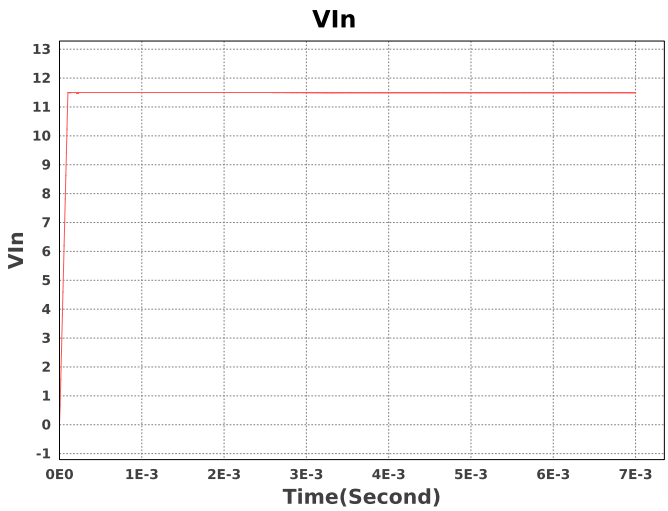
Simulation Type = Startup



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Rload	R	Load Resistance	51.42857142857143 ohm





Design Assistance

1. Master key : 3DA890648EAF22B9[v1]
2. **TPS55340** Product Folder : http://www.ti.com/product/TPS55340 : contains the data sheet and other resources.

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